Introduction to Structural Equation Modeling

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Contents

Course Information

In order to test a theory, we must express the theory as a statistical model and then test this model on quantitative (numeric) data. In this course we will use datasets from different disciplines within the social sciences (educational sciences, psychology, and sociology) to explain and illustrate theories and practices that are used in all social science disciplines to statistically model social science theories.

This course uses existing tutorial datasets to practice the process of translating verbal theories into testable statistical models. If you are interested in the methods of acquiring high quality data to test your own theory, we recommend following the course *Conducting a Survey* which is taught from November to January.

Most information about the course is available in this GitBook. Course-related communication will be through https://uu.brightspace.com (Log in with your student ID and password).

Acknowledgement

This course was originally developed by dr. Caspar van Lissa. I (dr. Kyle M. Lang) have modified Caspar's original materials and take full responsibility for any errors or inaccuracies introduced through these modifications. Credit for any particularly effective piece of pedagogy should probably go to Caspar.

- You can view the original version of this course here on Caspar's GitHub page.

Instructors

Coordinator:

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Lectures:

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Course overview

This course comprises three parts:

1. *Path analysis*: You will learn how to estimate complex path models of observed variables (e.g., linked linear regressions) as structural equation models.
2. *Factor analysis*: You will learn different ways of defining and estimating latent (unobserved) constructs.
3. *Full structural equation modeling*: You will combine the first two topics to estimate path models describing the associations among latent constructs.

Your mastery of these themes will be evaluated with a two data analysis assignments.

1. The first assignment covers path analysis.
    - You will complete this assignment in a group of three students.
    - This assignment will be graded on a pass/fail basis.
2. The second assignment covers latent variable modeling.
    - This is an individual assignment.

- Your course grade will be based on your second assignment grade.

Schedule

Course Week	Calendar Week	Lecture/Practical Topic	Workgroup Topic	Assignment Deadline
0	36	Pre-course preparation		
1	37	Introduction to R		
2	38	Statistical modeling, Path analysis		
3	39	Mediation, Moderation		
4	40	Confirmatory factor analysis (CFA) I	A1 Peer-Review	A1: 2025-10-05 @ 23:59
5	41	CFA II		
6	42	No class meetings		
7	43	Structural equation modeling (SEM)		
8	44	Multiple group models	A3 Peer-Review	
9	45	No class meetings		A3: 2025-11-02 @ 23:59

NOTE: The schedule (including topics covered and assignment deadlines) is subject to change at the instructors' discretion.

Learning goals

In this course you will learn how to translate a social scientific theory into a statistical model, how to analyze your data with these models, and how to interpret and report your results following APA standards.

After completing the course, you will be able to:

1. Translate a verbal theory into a conceptual model, and translate a conceptual model into a statistical model.
2. Independently analyze data using the free, open-source statistical software R.
3. Use a *path model* to represent the hypothesized causal relations among several variables, including relationships such as mediation and moderation.
4. Apply a *latent variable model* to a real-life problem wherein the observed variables are only indirect indicators of an unobserved construct.
5. Explain to a fellow student how *structural equation modeling* combines latent variable models with path models and the benefits of doing so.
6. Reflect critically on the decisions involved in defining and estimating structural equation models.

Background knowledge

We assume you have basic knowledge about multivariate statistics before entering this course. You do not need any prior experience working with R. If you wish to refresh your knowledge, Andy Field's *Discovering Statistics using R* is quite a popular and well-respected introductory stats book.

If you cannot access the Field book, many other introductory statistics textbooks cover these topics equally well. So, use whatever you have lying around from past statistics courses. You could also try one of the following open-access options:

- *Applied Statistics with R*
- *Introduction to Modern Statistics*
- *Introduction to Statistical Learning*

Resources

Literature

You do not need a separate book for this course!

Most of the information is contained within this GitBook and the course readings (which you will be able to access via links in this GitBook).

All literature is freely available online, as long as you are logging in from within the UU-domain (i.e., from the UU campus or through an appropriate VPN). All readings are linked in this GitBook via either direct download links or DOIs.

If you run into any trouble accessing a given article, searching for the title using Google Scholar or the University Library will probably due the trick.

Reading questions

Along with every article, we will provide reading questions. You will not be graded on the reading questions, but it is important to prepare the reading questions before every lecture. The reading questions serve several important purposes:

- Provide relevant background knowledge for the lecture
- Help you recognize and understand the key terms and concepts
- Make you aware of important publications that shaped the field
- Help you extract the relevant insights from the literature

Software

You will do all of your statistical analyses with the statistical programming language/environment R and the add-on package `lavaan`.

- If you want to expand your learning, you can follow this optional `lavaan` tutorial.

Logistics

Weekly preparation

Before every class meeting (both lectures and practicals) you need to do the assigned homework (delineated in the GitBook chapter for that week). This course follows a *flipped classroom* procedure, so you must complete the weekly homework to meaningfully participate in, and benefit from, the class meetings.

Evaluation

Your grade for the course is based on a "portfolio" composed of the two data analysis projects:

1. Path modeling
    - Deadline: Sunday 2025-10-05 at 23:59
    - Group assignment
    - Pass/Fail
2. Structural equation modeling
    - Deadline: Sunday 2025-11-02 at 23:59
    - Individual assignment
    - Comprises your entire numeric course grade

The specifics of the assignments will be explicated in the Assignments chapter of this GitBook

Attendance

Attendance is not mandatory, but we strongly encourage you to attend all lectures and practicals. In our experience, students who actively participate tend to pass the course, whereas those who do not participate tend to drop out or fail. The lectures and practicals build on each other, so, in the unfortunate event that you have to miss a class meeting, please make sure you have caught up with the material before the next session.

Using GenAI

Utrecht University (UU) has a set of general policies and guidelines regarding GenAI use in UU courses. Yet, the specifics are largely left for the instructors to decide on a course-by-course basis. So, in this section, we describe how you may use GenAI in this course.

AI index

UU has adopted the *AI Index* which defines five different "scenarios" for GenAI usage in UU courses. In the Faculty of Social and Behavioral Sciences (FSBS), we're piloting a revised version of the AI Index that delineates the following five scenarios.

Scenario A: No AI

The assessment is completed entirely without AI assistant in a controlled environment, ensuring that students rely solely on their existing knowledge, understanding, and skills.

You must not use AI at any point during the assessment. You must demonstrate your own core skills and knowledge.

Scenario B: AI planning

AI may be used for pre-task at activities such as brainstorming, outlining, and initial research. This scenario focuses on effective use of AI for planning, synthesis, and ideation, but assessment should emphasize the ability to develop and refine these ideas independently.

You may use AI for planning, idea development, and research. Your final submission should show how you have developed and refined these ideas.

Scenario C: AI collaboration

AI may be used to help complete the task including idea generation, drafting, feedback, and refinement. Students should critically evaluate and modify their AI suggested outputs, demonstrating their understanding.

You may use AI to assist with specific tasks such as drafting text, refining, and evaluating your work. You must critically evaluate and modify any AI generated content you use.

Scenario D: Full AI

AI may be used to complete any elements of the task, with students directing AI to achieve the assessment goals. In this scenario assessment may also require engagement with AI to achieve goals and solve problems.

You may use AI throughout your work either as you wish, or as specifically directed in your assessment. Focus on directing AI to achieve your goals while demonstrating your critical thinking.

Scenario E: AI exploration

AI is used creatively to enhance problem solving generate novel insights, what developed innovative solutions to solve problems. Students and educators Co design assessments to explore unique AI applications within the field of study.

You should use AI creatively to solve the task, potentially co- designing new approaches with your instructor.

Acceptable usage in this course

This course falls into Scenario C of the FSBS AI Index.

Why not A or B?

We don't see any point in being overly restrictive, and we think GenAI can be truly useful for rigorous data analysis. We want you to learn path analysis, factor analysis, and structural equation modeling, but we don't really care about the exact tools you use to do these analyses. Furthermore, data analysis and technical writing involve a lot of tedious grunt work. Your ability to complete these tasks tells us nothing about your mastery of the course content. So, we think it's great if GenAI obviates some of the menial drudgery and allows you to focus on the important work of developing, estimating, and interpreting statistical models. Consequently, Scenarios A and B are limiting and counterproductive.

Why not D or E?

You need to learn the underlying *why* of statistical modeling, not just the superficial *how*. In the era of modern statistical software packages, it's never been terribly difficult to estimate a statistical model, and that's even more true now that GenAI tools can help specify the function calls. The difficult part of statistical modeling has always been the critical reasoning needed to design a suitable analysis, evaluate the results, and synthesize the findings. You can't develop these critical reasoning skills if you offload too much work to GenAI. So, Scenario D works against our course goals.

Finally, this isn't a "vibe coding" course, we won't teach you how to use AI tools, and we don't care how good you are at using any given AI tool. So, Scenario E is clearly inappropriate

Including AI-generated content in your work

You are free to use GenAI to brainstorm ideas, draft text, prototype code, etc. However, you are always responsible for your own work, and *you are never allowed to submit work developed entirely by GenAI as your own*. In nearly all cases, you should adapt the robots' work to suit your project. For example,

Edit Text...

- Into your own voice
- For continuity and contextual fit
- To correct factual errors

Edit Code...

- To suit your analysis/project
- To use appropriate style
- For efficiency

Although you shouldn't need to synthesize any data for this course, be aware of the following restrictions:

- You may use AI tools to assist you in generating code that results in reproducible data sets.
- You may not use AI tools to directly generate data sets.

Referencing and academic integrity

You must document your use of GenAI tools and cite direct quotations of AI-generated content as dictated by the relevant UU policy. In your two data analysis assignments, you will also need to provide an AI disclosure statement.

Generative AI may be shiny new tech, but that just means AI is like the newest, shiniest gizmo in a toolbox full of time-tested options, not some paradigm shift in carpentry that deprecates the hammer. The normal ethics of scientific communication still apply:

- Transparently describe your work
- Honestly attribute credit/blame

Of course, the natural corollary of the above is that all the standard penalties for bad behavior also apply. Obfuscating your use of AI tools or attempting to claim AI-generated content as your own work is a form of academic fraud that is more-or-less equivalent to plagiarism. We will treat any such attempted subterfuge as academic dishonesty.

Finally, don't trust The Robots too much! You are ultimately responsible for the work you submit. So, you will have to own any "mistakes" that wily AI models induce in your work.

Assignments

This chapter contains the details and binding information about the two data analysis assignments that comprise the portfolio upon which your course grade is based.

Below, you can find a brief idea of what each assignment will cover.

- For each assignment, you will use R to analyze some real-world data, and you will write up your results in a concise report (not a full research paper).
  - Guidelines for these analyses/reports are delineated in the following two sections.
  - You will submit your reports via Brightspace.
- You will complete Assignment 1 in the *Assignment Group* that you join on Brightspace.
- You will complete the second assignment individually.
- The first assignment is graded as pass/fail.
  - You must pass this first assignment to pass the course.
- Your grade on the second assignment constitutes your numeric course grade.

Assignment 1: Path Analysis

For the first assignment, **you will work in groups** to apply a *path model* that describes how several (observed) variables could be causally related.

Required Components

The obligatory aspects of the first assignment are described below.

1. Choose a suitable dataset, and describe the data.
   - You can use any of the 8 datasets linked below.
   - Provide some form of citation for the data source
   - Provide descriptive statistics for important demographics (e.g., sex, age, SES) and the substantive variables that you will use to estimate your model.
2. State the research question; define and explicate the theoretical path model.
   - This model must include, at least, three variables.
     1. Dependent variable (DV)
     2. Theoretically interesting independent variable/predictor (IV)
     3. Covariate/control variable
   - Your model can also include more than one of any type of variable.
     - We encourage using more than one IV and covariate.
   - Include a path diagram that represents your theoretical model.
   - Explain the conceptual fit between your theory and your model.
     - I.e., how does your model represent your theory?
3. Translate your theoretical path model into `lavaan` syntax, and estimate the model.
   - Submit the (clean) R script along with your report.
     - Submit this R script as a separate file, not part of your report.
   - Include the lavaan syntax used to estimate your path models in your report.
     - Place these syntax snippets in an appendix.
4. Report the results of you analysis.
   - Evaluate the model assumptions.
   - Provide relevant output in a suitable format.
   - Include measures of explained variance for the dependent variables.
5. Discuss your findings.
   - Use your results to answer the research question.

- Consider the strengths and limitations of your analysis.
- Discuss any important decisions that could have influence your results.

Evaluation

See the Grading section below for more information on how Assignment 1 will be evaluated.

You can access an evaluation matrix for Assignment 1 here.

- This matrix gives an indication of what level of work constitutes *insufficient*, *sufficient*, and *excellent* responses to the five components described above.

Submission

Assignment 1 is due at 23:59 on Sunday 5 October 2025.

- Submit your report via the *Assignments Page* on Brightspace.

Assignment 2: Structural Equation Modeling

In the second assignment, **you will work individually** to apply a structural equation model (SEM) that describes how several (latent) variables could be causally related.

Before estimating your SEM, you will evaluate the validity of your latent variables by using confirmatory factor analysis (CFA) to test your hypothesized measurement model. You will include both your CFA and SEM results in your write-up. The inferences that you draw from an SEM are only valid if the underlying measurement model holds. So, the first step in reporting any SEM (at least any SEM that includes estimated latent constructs) is demonstrating the validity of the measurement model by reporting your CFA results.

Required Components

The obligatory aspects of the second assignment are described below.

1. Choose a suitable dataset, and describe the data.
    - Unless you have a good reason to switch, you should work with the same data that you analyzed in Assignment 1.
    - If you need to switch, you can use any of the 8 datasets linked below.
2. State the research question; define and explicate the theoretical SEM.
    - The structural component of this model must include, at least, three variables.
    - The structural model must include, at least, two latent variables.
    - At least one independent variable in your model must be latent.
    - You do not need to explicate a separate theory/research question/hypothsis for your measurement model.
3. Translate your hypothesized measurement model into **lavaan** syntax, and estimate the CFA model.
    - Submit the (clean) R script along with your report.
        - Submit this R script as a separate file, not part of your report.
    - Include the lavaan syntax used to estimate your CFA models in your report.
        - Place these syntax snippets in an appendix.
4. Report the results of your CFA.
    - Provide relevant output in a suitable format.
    - Include suitable parameter estimates.
    - Include measures of explained variance for the observed indicators.
    - Include measures of model fit.
    - Evaluate the model assumptions.
5. Translate your theoretical SEM into **lavaan** syntax, and estimate the model.
    - Submit the (clean) R script along with your report.
        - Submit this R script as a separate file, not part of your report.
    - Include the lavaan syntax used to estimate your SEM model in your report.
        - Place these syntax snippets in an appendix.
6. Report the results of your SEM.
    - Provide relevant output in a suitable format.
    - Include suitable parameter estimates.
    - Include measures of model fit.

- Include measures of explained variance for the latent dependent variables.

7. Discuss your findings.
    - Use your results to answer the research question.
    - Consider the strengths and limitations of your analysis.
    - Discuss any important decisions that could have influence your results.

Evaluation

See the Grading section below for more information on how Assignment 2 will be evaluated.

You can access an evaluation matrix for Assignment 2 here.

- This matrix gives an indication of what level of work constitutes *insufficient*, *sufficient*, and *excellent* responses to the seven components described above.

Submission

Assignment 2 is due at 23:59 on Sunday 2 November 2025.

- Submit your report via the *Assignments Page* on Brightspace.

Elaboration & Tips

Progression

The two assignments are meant to build upon one another. So, you should plan on analyzing the same dataset for both assignments and choose a theoretical model that will satisfy the requirements of both assignments. To facilitate this progression, consider the following points.

- For A2, your theoretical model must include two or more latent constructs. At least one of the latent constructs must act as in IV in your model.
    - When specifying your model for A1, ensure that you use at least two scale scores constructed from three or more items each.
    - Make sure to include at least one of these scale scores as an IV.
    - In A2, you can use latent variables to represent these scales scores.
- We will not cover methods for estimating latent variables from categorical items.
    - When specifying the latent constructs/scale scores in your model, do not use nominal items and try to avoid ordinal items with fewer than five levels.
- We will not cover methods for latent variable interactions.
    - When specifying your theoretical model in A1, do not include any interactions involving scale scores (i.e., variables that will be latent constructs in A2).
    - *EXCEPTION:* You may specify an interaction between a scale score/latent construct and a nominal grouping variable (e.g., sex, nationality, ethnicity).
        - You will learn how to test this type of hypothesis via multiple-group SEM in Week 8.

Recycling Prior Work

Since A2 builds on A1, you'll inevitably find yourself in the position of needing to document exactly the same things across assignments. In such circumstances, you'll undoubtedly be wondering if you can reuse some of your previous writing.

Yes. You're free to reuse parts of your reports across all three assignments. There are some minor caveats to this statement, however.

- When reusing prior work, make sure to address any feedback that you received on that work.
    - Uncorrected mistakes in copied passages will incur especially harsh penalties in the grading process.
- Make sure to update any recycled passages to suit the most recent analyses.
    - A path diagram from A1 won't be valid for A2.
    - A description of your theoretical model from A1 that refers to "scale scores", won't be valid for A2.
- Don't keep irrelevant parts of past assignments.
    - Your grade will be penalized for including extraneous information.

Theoretical Model & Research Question

You need to provide some justification for your model and research question, but not too much. Only provide enough to justification to demonstrate that you've actually conceptualized and estimated a theoretically plausible statistical model (as opposed to randomly combining variables until **lavaan** returns a pretty picture).

- You have several ways to show that your model is plausible.
  - Use common-sense arguments.
  - Reference (a small number of) published papers.
  - Replicate an existing model/research question.
- Don't provide a rigorous literature-supported theoretical motivation.
  - You don't have the time to conduct a thorough literature review, and we don't have the time to read such reviews when grading.
  - Literature review is not one of the learning goals for this course, so you cannot get "bonus points" for an extensive literature review.

You are free to test any plausible model that meets the size requirements.

- You can derive your own model/research question or you can replicate a published analysis.

Model Specifications

We will not cover methods for modeling categorical outcome variables. So, use only continuous variables as outcomes.

- DVs in path models and the structural parts of SEMs
- Observed indicators of latent factors in CFA/SEM

NOTE: You may treat ordinal items with five or more levesl as continuous, for the purposes of these assignments.

We will not cover methods for latent variable interactions.

- Don't specify a theoretical model that requires an interaction involving a latent construct.

There is one exception to the above prohibition. If the moderator is an observed grouping variable, you can estimate the model as a multiple-group model. We'll cover these methods in Week 8.

Assumptions

You need to show that you're thinking about the assumptions and their impact on your results, but you don't need to run thorough model diagnostics. Indeed, the task of checking assumptions isn't nearly as straight forward in path analysis, CFA, and SEM as it is in linear regression modeling. You won't be able to directly apply the methods you have learned for regression diagnostics, for example.

Since all of our models are estimated with normal-theory maximum likelihood, the fundamental assumption of all the models we'll consider in this course boils down to the following.

> All random variables in my model are i.i.d. multivariate normally distributed.

So, you can get by with basic data screening and checking the observed random variables in your model (i.e., all variables other than fixed predictors) for normality.

- Since checking for multivariate normality is a bit tricky, we'll only ask you to evaluate univariate normality.
- You should do these evaluations via graphical means (e.g., Normal QQ-Plots).

To summarize, we're looking for the following.

Data

- Consider whether the measurement level of your data matches the assumptions of your model.
- Check your variables for univariate outliers.
  - If you find any outliers, either treat them in some way or explain why you are retaining them for the analysis.
- Check for missing data.
  - For the purposes of the assignment, you can use complete case analysis to work around the missing data.
  - If you're up for more of a challenge, feel free to try multiple imputation or full information maximum likelihood.

Model

- Evaluate the univariate normality of any random, observed variables in your model.
  - E.g., DVs in path models, observed IVs modeled as random variables, indicators of latent factors
  - If you fit a multiple-group model for Assignment 2, do this evaluation within groups.
  - Use graphical tools to evaluate the normality assumption.
    - Normal QQ-Plots
    - Histograms
    - Kernel Density Plots

Reporting Standards

What do we mean by reporting your results "in a suitable format"? Basically, put some effort into making your results readable, and don't include a bunch of superfluous information. Part of demonstrating that you understand the analysis is showing that you know which pieces of output convey the important information.

- Tabulate your results; don't directly copy the R output into your report.
- Don't include everything **lavaan** gives you.
- Include only the output needed to understand your results and support your conclusions.

The purpose of statistical data analysis is to provide empirical evidence for or against some theory/hypothesis/model. Therefore, the way you document your analysis and report your results must serve this basic purpose, first and foremost.

- Support any potentially refutable claims with appropriate statistics from your analysis or a suitable citation.
- Not every statement needs supporting evidence.
  - Scientific laws
  - Physical constants
  - Logical (in the technical sense) implications of irrefutable antecedents
  - Your own opinions (assuming you're not trying to pass them off as facts)

The next three subsections cover some specific considerations for reporting common classes of statistical results.

Significance Tests

Any claim of a significant effect must be supported by relevant statistical evidence.

- Match your test to your hypothesis
  - Directional hypothesis $\Rightarrow$ one-tailed test
  - Hypothesis of any non-zero effect $\Rightarrow$ two-tailed test
- For any test of an estimated parameter (e.g., mean, mean difference, regression coefficient, covariance), report the parameter estimate.
- When using a test statistic (e.g., t, z, F, χ^2) to conduct the test, report:
  - The estimated test statistic
  - The degrees of freedom for the test statistic
  - The p-value for the test statistic
    - When $p \geq 0.001$, report the computed p-value rounded to three decimal places.
    - Otherwise, report the p-value as $p < 0.001$.
- When using a confidence interval to conduct the test:
  - Clearly indicate the confidence level used to define the interval
  - For directional hypotheses/one-tailed tests, only report the relevant interval bound, and report the other bound as $\pm\infty$.
- When using χ^2 difference tests for significance testing, report
  - The $\Delta\chi^2$ statistic
  - The Δdf
  - The p-value for the $\Delta\chi^2$

Model Fit

Judging model fit is always a subjective process. The key to success is providing multiple pieces of convergent evidence to support the claimed degree of fit.

- Always report the χ^2, its degrees of freedom, and the associated p-value.
  - Don't interpret the χ^2: it's testing an uninteresting hypothsis that we don't expect to hold.
- Report at least two additional, non-redundant fit indices (i.e., indices that quantify fit in different ways).

- If you don't have a particular preference, I'd recommend CFI, RMSEA (and its 90% CI), and SRMR

Parameter Estimates

Obviously, you need to report any parameter estimates that directly represent some component of your theory (e.g., regression coefficients that quantify linear associations implied by your theory). You also need to report the significance tests for these parameters.

When evaluating a measurement model, a few key parameter matrices come into play. In addition to showing that your CFA model adequately fits the data, you should report the following parameter estimates:

- Latent variances and covariances
- Factor loadings
- Residual variances

If your CFA includes a mean structure, you should also report any estimated latent means and item intercepts.

Data

Below, you can find links to a few suitable datasets that you can use for the assignments.

- You *must* use one of the following datasets. You may not choose your own data from the wild.

Coping with Covid

- Dataset
- Codebook
- Pre-Registration

Feminist Perspectives Scale

- Dataset
- Article

Hypersensitive Narcissism Scale & Dirty Dozen

- Dataset
- HSNS Article
- DD Article

Kentucky Inventory of Mindfulness Skills

- Dataset
- Article

Depression Anxiety Stress Scale

- Dataset
- DASS Information

Nomophobia

- Dataset

Recycled Water Acceptance

- Dataset
- Article

Procedures

Formatting

Your assignment submissions must follow certain formatting guidelines. Failing to correctly format your submissions can impact your grade.

Report

You must submit your Assignment 1 & 2 reports in PDF format.

- Each report must include the following information:
  - The names of all assignment authors (i.e., all group members for Assignment 1, your name for Assignment 2).
  - The Assignment Group number (only for Assignment 1).

Your reports should include appendicies containing the lavaan syntax used to estimate each model include in the respective report.

- Include only the lavaan syntax string and the call to the estimation function used to fit the model (e.g., `cfa()`, `sem()`, `lavaan()`).

For example, the following snippets would be appropriate.

```
cfaMod <- '
f1 =~ a1 + a2 + a3
f2 =~ b1 + b2 + b3
'

cfaOut <- cfa(cfaMod, data = dat1, std.lv = TRUE)
```

```
semMod <- '
f1 =~ a1 + a2 + a3
f2 =~ b1 + b2 + b3

f1 ~ f2 + x
'

semOut <- sem(semMod, data = dat1, std.lv = TRUE)
```

The snippet below, however, contains additional unecessary commands related to data ingest/processing and summarizing the model.

```
dat0 <- readRDS("../data/dataset.rds")
dat1 <- rename(dat0, x = foo, y = bar, z = baz)

pathMod <- 'foo ~ bar + baz'
pathOut <- sem(pathOut, data = dat1)

summary(pathOut)
```

Syntax File

You must also submit an executable script containing the code used for your analyses.

- If you used **KnitR** to embed your analysis code in your report, you may submit the source code that generates your report (i.e., an RMD [R Markdown], QMD [Quarto], or RNW [LaTeX] file).
- Otherwise, submit a standard R script.

Clean the script before submitting.

- Remove extraneous/redundant code and comments.
  - Only include the code for analyses that appear in your report.
  - Include code for data processing/clean/wrangling.
- Order the commands to match your final analysis.
  - E.g., Data ingest → Data processing → Descriptive analysis → Modeling → Processing results → Post-hoc analysis

    - In theory, I should be able to source your script (i.e., run the entire script in "batch mode") to reproduce your results.

Length

You may use as many words as necessary to adequately explain yourself, but concision and parsimony are encouraged. Note that the assignments are **not** intended to be full-blown research papers! The focus should be on the definition of your model, how this model relates to theory (introduction), and what you have learned from your estimated model (discussion).

For both assignments, you should be able to get the job done in fewer than 10 pages of text (excluding title page, figures, appendices, and references).

Submission

You will submit your reports through Brightspace.

- Each assignment has a corresponding item in the "Assignments" section of the Brightspace page through which you will submit your reports.
- For Assignment 1, you may only submit one report per group.
    - Designate one group member to submit the report.
    - The grade for this submission will apply to all group members.
- If something goes wrong with the submission, or you notice a mistake (before the deadline) that you want to correct, you may upload a new version of your report.
    - We will grade the final submitted version.
- The submissions will be screened with *TurnItIn* to check for plagarism.

Grading

Group Assignments

Assignment 1 is simply graded as pass/fail. To pass, you must:

1. Make a reasonable effort to complete all required work
    - Include all required components listed above in your A1 report
2. Demonstrate a satisfactory ability to apply path analysis
    - Does your work demonstrate that you understand what you're doing?
        - Mindlessly implementing a sequence of memorized actions is not sufficient.
    - Does your report show that you understand what you've done, why you've done it, and the implications of your results?
    - Can we understand your analysis and results after reading your report?
3. Submit your assignment before the deadline

Otherwise, you will fail the assignment.

Individual Assignment

Assignment 2 will be fully graded on the usual 10-point scale. We will follow the procedure described below to compute your A2 grade.

- Each report starts with a baseline grade of 7.
- We will evaluate the quality of your work to adjust your grade above or below this baseline.
    - As we read through the report, we add credit (e.g., 0.1 points, 0.25 points, 0.5 points) for each "good" aspect that demonstrates above-average understanding or effort.
    - Likewise, we deduct credit for each "bad" aspect that indicates below-average understanding or effort.

So, if your report aligns with a satisfactory (but not exceptional) level of understanding and effort, your grade will be 7. Your mark will be higher, however, to the extent that you can demonstrate more-than-satisfactory levels of mastery and/or effort. Likewise, your grade will be lower if you do not meet expectations. Importantly, the added and deducted credit will be assigned independently, so the additional points you get for good aspects can counteract any points you may lose for negative aspects.

- The adjustment points will be allocated according to the extent to which your submission addresses the required components listed above.

- The evaluation matrix gives an indication of how these points will be apportioned.

Assuming your group passes the first assignment, your final course grade will simply be your Assignment 2 grade.

Resits

You must get a "pass" for Assignment 1 and score at least 5.5 on Assignment 2 to pass the course. If you fail either of the assignments, you will have the opportunity to resit the failed assignment(s).

Procedure

The resit procedure is very simple. You revise the failed assignment, and your score on the revised submission will replace your failing grade (assuming a better grade for the revision).

Take note of the following points.

- The same submission deadline applies to both resit assignments. You are, of course, free to submit your resits early.
- You must resit each failed assignment. If, for example, you failed both A1 and A2, you must revise both A1 and A2 and submit both revisions before the deadline.
- You must complete the A1 resit in your original groups. If some group members have dropped the course, complete the A1 resit with the remaining group members.
- You must complete the A2 resit individually.
- Your revised grade for A2 cannot be higher than 6.
- Submit your resit assignments through Brightspace.

Grading Standards

Your resit score will not be a simple additive transformation of your original score. We will not grade the resits by comparing your original report to your resit and checking if you corrected "enough" issues to have produced a passing score in the context of the original report. Such an approach would be the educational testing equivalent of HARKing. So, if you score a 5 on your original submission, for example, and you correct two big issues (the absence of which would have implied a passing score in the original report), you can still fail the resit.

The resits are your second chance to demonstrate mastery of the relevant concepts (i.e., path analysis, latent variable modeling). If your corrections bring the report up to a level that seems to suggest that you "know what you are doing", you will pass. If your report still suggests fundamental misunderstandings (or is otherwise substantially deficient), you will not pass. A passing grade in this course is supposed to show that you can use lavaan to conduct a sensible path analysis, CFA, and SEM, not that you can hack some haggard instructor's goofy evaluation scheme.

I don't want any of you to fail this course. I derive no pleasure from your suffering. Anyway, from a purely selfish perspective, failing grades make much more work for the instructors than passing grades do. That being said, I do want the degree you earn from UU to be a meaningful marker of the skills and knowledge that your study program is meant to impart. So, if you did not achieve the learning goals for this course, you should not pass the course, even though that's an unpleasant outcome for all parties.

Example Assignment

You can find an example of a good submission (for an older version of Assignment 2) here.

This example is not perfect (no paper ever is), and several points could be improved. That being said, this submission exemplifies what we're looking for in your project reports. So, following the spirit of this example would earn you a high grade.

Rules

Resources

For both assignments, you may use any reference materials you like, including:

- All course materials
- The course GitBook
- Additional books and papers
- The internet

Collaboration

You will complete the first assignment in groups.

- Although you will work in groups, your group may not work together with other groups.

You will complete the final assignment individually.

- For this assignment, you may not work with anyone else.

For both assignments, you are obligated to submit original work (i.e., work conducted for this course by you or your group).

- Submitting an assignment that violates this condition constitutes fraud.
- Such cases of fraud will be addressed according to the University's standard policy.

Academic integrity

Hopefully, you also feel a moral obligation to obey the rules. For this course, we have implemented an examination that allows you to showcase what you have learned in a more realistic way than a written exam would allow.

- This assessment format spares you the stress of long exams (the two exams for this course used to be 4 hours each) and the attendant studying/cramming.
- The assignments will also help you assess your ability to independently analyse data, which is important to know for your future courses and/or career.

However, this format also assumes that you complete the assignments in good faith. So, I simply ask that you hold up your end of the bargain, and submit your original work to show us what you've learned.

GenAI disclosure statement

Naturally, all of the course policies on GenAI also apply to the data analysis assignments. In addition to these general policies, you must include a GenAI disclosure statement in each of your assignments.

You are free to use AI tools like OpenAI's ChatGPT or GitHub CoPilot, but you must cite the tools you used in the same way that you would cite any other software. As with citing any other source, properly citing AI tools both attributes the appropriate credit to the tool's developers and helps you clearly and accurately describe your own work. In addition to providing a valid citation, you must also explain how you used the AI tool. Specifically, did you use AI to:

1. Create new content (e.g., use ChatGPT to draft some parts of the text)
2. Modify content that you created (e.g., use CoPilot to debug your R code)

Provide this information in a separate *GenAI Disclosure Statement* section at the end of your report.

Strict stuff

By submitting your assignments (both group and individual), you confirm the following:

1. You have completed the assignment yourself (or with your group)
2. You are submitting work that you have written yourself (or with your group)
3. You are using your own UU credentials to submit the assignment
4. You have not had outside help that violates the conditions delineated above while completing the assignment

All assignments will be submitted via Ouriginal in Blackboard and, thereby, checked for plagiarism. If fraud or plagiarism is detected or suspected, we will inform the Board of Examiners in the usual manner. In the event of demonstrable fraud, the sanctions delineated in Article 5.15 of the Education and Examination Regulations (EER) will apply.